

Reference List: Query Composition Methods for GeoCIR

Each formulation is covered in two sections. **Section 1** collects papers that implement or directly evaluate the composition strategy in a CIR (or closely related retrieval) context. **Section 2** collects the theoretical ancestors — foundational works that introduced the underlying mathematical idea, even if they pre-date CIR or target a different downstream task entirely.

Formulation A — Linear Composition

$$q = \alpha \cdot E_I(I_r) + \beta \cdot E_T(T_g)$$

The idea is to produce a composed query by forming a weighted sum of the image and text embeddings. Operationally simple, training-free, and forms the baseline against which every more principled method is judged.

Section 1 — CIR-specific implementations

[A-CIR-1] Vo et al. "Composing Text and Image for Image Retrieval — An Empirical Odyssey." CVPR 2019. The paper that founded the CIR task. Alongside TIRG, it systematically benchmarks simpler alternatives including concatenation followed by a linear layer and direct element-wise sum, establishing them as official baselines. Every later paper that reports a "concat" or "sum" baseline traces back to this experimental setup.

[A-CIR-2] Couairon, Douze, Cord, Schwenk. "Embedding Arithmetic of Multimodal Queries for Image Retrieval." CVPRW 2022. The most direct CIR implementation of linear arithmetic. The method computes a delta vector from a text modifier and adds it to an image embedding in CLIP space, then retrieves by nearest-neighbor. The paper introduces SIMAT, a diagnostic benchmark for testing whether CLIP's embedding space supports delta-based image transformation. Key finding: vanilla CLIP embeddings support addition poorly, but light fine-tuning on COCO dramatically improves arithmetic faithfulness. This is the closest existing paper to Formulation A's spirit applied in a CLIP setting, and is directly relevant to assessing whether simple addition will work for GeoCIR.

[A-CIR-3] Park et al. "Spherical Linear Interpolation and Text-Anchoring for Zero-shot Composed Image Retrieval." ECCV 2024. Rather than naive Euclidean addition, this paper interpolates between the image and text embeddings using Spherical Linear Interpolation (Slerp) on the unit hypersphere. The α/β weighting in Formulation A implicitly lives on the chord between the two embeddings; Slerp follows the geodesic on the sphere instead, which is geometrically more appropriate for normalized CLIP embeddings. The paper also shows that aligning the image embedding distribution to the text embedding distribution via LoRA ("Text-Anchoring Training") before interpolation significantly improves retrieval. Together this provides both the interpolation variant of Formulation A and a recipe for making the two embeddings commensurable.

[A-CIR-4] Baldrati et al. "Effective Conditioned and Composed Image Retrieval Combining CLIP-based Features." CVPR 2022. [Combiner/CLIP4CIR] While the key contribution is a learned MLP combiner (Formulation D territory), the paper's ablation tables include element-wise addition and concatenation variants over CLIP features, providing quantitative evidence on how much linear composition underperforms a learned alternative — useful data for motivating the move to Formulations B through D.

Section 2 — Theoretical foundations

[A-TH-1] Mikolov, Chen, Corrado, Dean. "Efficient Estimation of Word Representations in Vector Space." ICLR 2013. The Word2Vec paper. First systematic demonstration that a continuous embedding space learned by a shallow neural network supports linear semantic arithmetic: "king – man + woman $\approx$ queen." Established the analogy paradigm that directly motivates Formulation A and B. The country-capital analogies (Paris – France + Germany $\approx$ Berlin) are structurally identical to the geographic pivot operation in Formulation B.

[A-TH-2] Mikolov, Yih, Zweig. "Linguistic Regularities in Continuous Space Word Representations." NAACL-HLT 2013. Companion paper to Word2Vec that formally introduces and analyzes the 3CosAdd method for analogy solving (find $\arg \max_b \cos(b, a^* - a + b^*)$). Provides the theoretical framing for the subtraction-and-addition paradigm. The caveats discussed here — that the source words contaminate the result, that the geometry works best for high-frequency, evenly-distributed concepts — are directly relevant to understanding when Formulation A/B will and will not work.

[A-TH-3] Pennington, Socher, Manning. "GloVe: Global Vectors for Word Representation." EMNLP 2014. Provides an alternative derivation of embedding arithmetic grounded in co-occurrence ratios. The GloVe paper's theoretical analysis shows that arithmetic works when the PMI difference between word pairs is constant across an analogy class. For GeoCIR, this implies Formulation A will work best when landmark pairs across geographic regions have consistent co-occurrence relationships with geographic tokens in the training corpus — a testable empirical claim.

[A-TH-4] Radford et al. "Learning Transferable Visual Models from Natural Language Supervision." ICML 2021. [CLIP] CLIP's contrastive training aligns image and text into a shared L2-normalized 512-D space. This is the embedding space that makes Formulation A executable: without a shared space, adding an image vector and a text vector produces a meaningless result. CLIP provides the prerequisite joint representation. The paper's zero-shot transfer results already demonstrate that simple similarity-based combination of visual and textual features works surprisingly well, foreshadowing Formulation A's viability.

[A-TH-5] Ethayarajh, Duvenaud, Hirst. "Towards Understanding Linear Word Analogies." ACL 2019. Formal analysis of when and why embedding arithmetic succeeds, grounded in a constant-shift PMI (csPMI) theorem: arithmetic works precisely when the PMI difference between pairs in an analogy class is constant. Provides the clearest theoretical account of Formulation A's failure modes and is useful background for understanding why Formulation B (GPS pivot) should be more reliable than naive addition.

Formulation B — GPS-Pivot Arithmetic

$$q = E_I(I_r) - E_L(\phi_r, \lambda_r) + E_T(T_g)$$

The distinctive feature of this formulation is that it explicitly uses the GPS embedding of the reference location as a "pivot" to remove the reference's geographic component before injecting the target geographic modifier. This is only possible because GeoCLIP aligns all three modalities — image, GPS, and text — in the same embedding space.

Section 1 — CIR-specific implementations

[B-CIR-1] Couairon et al. "Embedding Arithmetic of Multimodal Queries for Image Retrieval." CVPRW 2022. Cited again here because this paper is the closest CIR work to Formulation B in structure. It performs image + text delta arithmetic in CLIP space and explicitly discusses the geometry of delta vectors (direction and magnitude). The failure mode identified — that the geographic component already in the image embedding "resists" the text delta — is exactly the problem Formulation B's GPS subtraction step is designed to solve.

[B-CIR-2] Vivanco Cepeda, Nayak, Shah. "GeoCLIP: CLIP-Inspired Alignment between Locations and Images for Effective Worldwide Geo-localization." NeurIPS 2023. The enabling paper for Formulation B. Section 4.4 demonstrates the text-GPS alignment property: because the image encoder uses a frozen CLIP backbone, the location encoder also inherits alignment with CLIP's text features. The paper shows qualitatively that text queries such as "Desert" or "Snow" produce globally correct GPS similarity maps. Formulation B is a direct compositional application of this three-way alignment: if E_I , E_L , and E_T all co-reside in the same space, then GPS-pivot subtraction-and-addition is geometrically meaningful.

[B-CIR-3] Kim et al. "Dual Compositional Learning in Interactive Image Retrieval." AAAI 2021. [DCNet] While not using GPS, this paper is structurally important because it explicitly trains a Correction Network that models the *difference vector* between reference and target image: $I_t - I_r \rightarrow T_q$. The inverse direction — using a difference vector to represent a "relocation" — is conceptually identical to what Formulation B does when it subtracts $E_L(\phi_r, \lambda_r)$ from $E_I(I_r)$ to extract a location-free visual residual. DCNet provides empirical evidence that difference vectors are learnable and meaningful in composed retrieval.

Section 2 — Theoretical foundations

[B-TH-1] Mikolov et al. Word2Vec (ICLR 2013) and Linguistic Regularities (NAACL 2013). See [A-TH-1] and [A-TH-2] above. Formulation B is structurally a geographic analogy: "Tokyo Skytree is to Japan as X is to Germany." The 3CosAdd analogy format maps directly: $X \approx E_I(\text{Tokyo Skytree}) - E_L(\text{Japan}) + E_T(\text{Germany})$.

[B-TH-2] Vivanco Cepeda et al. GeoCLIP (NeurIPS 2023). Listed again as a theoretical foundation because the GPS encoder architecture is the mathematical tool that enables

Formulation B. Specifically, the location encoder uses $M=3$ scale Random Fourier Features with learnable frequency matrix R and 4-layer MLP per scale, producing a 512-D GPS embedding trained via contrastive loss against images in the same shared space. Understanding this architecture is necessary for understanding why $E_L(\phi_r, \lambda_r)$ is a valid entity to subtract from $E_I(I_r)$.

[B-TH-3] Mai, Janowicz, Yan, Zhu, Regalia, Cai. "Multi-Scale Representation Learning for Spatial Feature Distributions Using Grid Cells." ICLR 2020. [Space2Vec] Introduces the multi-scale sinusoidal encoding for geographic coordinates, directly used by GeoCLIP's location encoder design. The key insight — that hierarchical spatial frequency encoding using multiple σ values captures both fine-grained and coarse-grained geographic structure — grounds the claim that GPS embeddings respect geographic proximity and continuity across scales, which is a prerequisite for the subtraction in Formulation B to remove a spatially localized "Japan" signal.

[B-TH-4] Yin, Liu, Zheng, Jensen, Gartner. "GPS2Vec: Towards Generating Worldwide GPS Embeddings." SIGSPATIAL 2019 / IEEE TMM 2021. Precursor to GeoCLIP's location encoder. GPS2Vec learns 512-D semantic embeddings for GPS coordinates from geotagged Flickr documents by treating co-occurring photos as contextually related locations. Demonstrates that GPS coordinates can be embedded in a semantic space where nearby coordinates have similar vectors — the foundational claim that makes $E_L(\phi_r, \lambda_r)$ subtractable.

[B-TH-5] Klemmer, Rolf, Robinson, Mackey, Regalia, Subramanian. "SatCLIP: Global, General-Purpose Location Embeddings with Satellite Imagery." AAAI 2025. A direct successor to GeoCLIP that uses Sentinel-2 satellite imagery and spherical harmonics + SIREN to learn location embeddings. Relevant because it provides a richer, higher-resolution alternative location encoder that could replace E_L in Formulation B. If GeoCLIP's location encoder proves too coarse for fine-grained city-level geographic conditioning, SatCLIP offers a drop-in alternative.

Formulation C — Geometric Projection (Visual-Geographic Decomposition)

$$q = [E_I(I_r) - \text{proj}_G(E_I(I_r))] + E_T(T_g)$$

This formulation does not require GPS coordinates at inference time. Instead, it decomposes the image embedding by projecting onto a pre-computed geographic subspace G and subtracting the projection, leaving a geographic-agnostic visual residual. The geographic subspace is computed offline via PCA over the GPS encoder's output.

Section 1 — CIR-specific implementations

[C-CIR-1] Lee, Kim, Han. "CoSMo: Content-Style Modulation for Image Retrieval with Text Feedback." CVPR 2021. The most structurally analogous CIR method to Formulation C. CoSMo explicitly disentangles the image embedding into a content branch (preserved) and a style branch (modified by text). The style branch is conceptually equivalent to the "geographic component" in Formulation C: the text modifier is routed into the part of the embedding that the modification should affect, while the content (visual structure, architecture type) is held fixed.

CoSMo provides empirical validation that such disentanglement improves retrieval over methods that treat the full image embedding as undifferentiated.

[C-CIR-2] Baldrati et al. "Zero-Shot Composed Image Retrieval with Textual Inversion." ICCV 2023. [SEARLE/iSEARLE] Relevant here because the SEARLE optimization procedure can be interpreted as projecting the image embedding into the CLIP text embedding subspace, then combining with the modification text. The concept of finding the nearest point in the text token space to a given image embedding is a special case of projection onto a target subspace.

[C-CIR-3] Jia et al. "Towards Interpretable Geo-localization: a Concept-Aware Global Image-GPS Alignment Framework." arXiv:2509.01910, 2025. Directly relevant: extends GeoCLIP with a concept bank that projects image and GPS embeddings onto named geographic concepts ("tropical climate," "cathedral," "forest"). The concept bank provides an explicit characterization of the geographic subspace G in Formulation C — rather than deriving G via unsupervised PCA over GPS embeddings, the concept bank defines G via textual geographic concepts. The UMAP visualizations in this paper showing text, image, and GPS forming a coherent shared space are the closest published evidence for the geometric structure that Formulation C exploits.

Section 2 — Theoretical foundations

[C-TH-1] Hotelling. "Analysis of a Complex of Statistical Variables into Principal Components." Journal of Educational Psychology, 1933. / Pearson. "On Lines and Planes of Closest Fit to Systems of Points in Space." Philosophical Magazine, 1901. [PCA] Principal Component Analysis is the offline computation at the core of Formulation C. The pre-computation of the geographic subspace G from a set of GPS embeddings $\{E_L(\varphi_i, \lambda_i)\}$ is a standard PCA problem: the top- k principal components of the GPS embedding matrix define the projection matrix P_G . Knowing that PCA recovers the directions of maximum variance in the GPS embedding cloud provides the justification for why $\text{proj}_G(E_I(I_r))$ extracts the geographically informative dimensions from the image embedding.

[C-TH-2] Mikolov, Jouit, Chen, Dean, Corrado. "Exploiting Similarities among Languages for Machine Translation." arXiv:1309.4168, 2013. Introduces the idea of learning a linear mapping between embedding spaces of different languages by aligning their shared semantic structure. The approach demonstrates that a linear projection can successfully isolate language-specific information from semantic content — the conceptual predecessor to Formulation C's geographic projection. If this works for language pairs, the same principle should apply to the geographic vs. visual-semantic decomposition in GeoCLIP's shared space.

[C-TH-3] Bolukbasi, Chang, Zou, Saligrama, Kalai. "Man is to Computer Programmer as Woman is to Homemaker? Debiasing Word Embeddings." NeurIPS 2016. Introduces the subspace projection approach to removing a specific component (gender) from word embeddings: define a "gender subspace" from a set of definitionally gendered word pairs, then project all embeddings onto the orthogonal complement. This is structurally identical to Formulation C's geographic decontextualization: define a "geographic subspace" from GPS embeddings, project the image embedding onto the complement, and the result is the geographic-agnostic visual residual. The paper provides clear mathematical exposition and

empirical evidence that subspace projection successfully removes a semantic attribute from embeddings.

[C-TH-4] Vivanco Cepeda et al. GeoCLIP (NeurIPS 2023). The GPS encoder output distribution defines the geographic subspace G . Formulation C's pre-computation step relies on sampling from this distribution (i.e., evaluating E_L over a dense grid of GPS coordinates) and applying PCA.

Formulation D — Textual Inversion

$$p_r = P(E_I(I_r)) ; q = E_T("[p_r] \text{ located in } T_g")$$

A learned projection network P maps the image embedding to a continuous pseudo-token in CLIP's text embedding space. The composed query is formed by inserting this pseudo-token into a geographic template and passing the hybrid sequence through the frozen CLIP text encoder.

Section 1 — CIR-specific implementations

[D-CIR-1] Saito, Matsumoto, Panda, Chang, Vedantam, Feris, Dreyer, Saenko. "Pic2Word: Mapping Pictures to Words for Zero-Shot Composed Image Retrieval." CVPR 2023. The founding paper of the textual inversion paradigm for ZS-CIR. Trains a 3-layer MLP projection $P: \mathbb{R}^{512} \rightarrow \mathbb{R}^{512}$ that maps image embeddings to pseudo-tokens, supervised only by a loss that brings $P(E_I(I))$ close to $E_T(\text{caption of } I)$ from CC3M image-caption pairs. At inference, the composed query is $E_T("[p_r] \text{ and } T_m")$. The direct template "a $[p_r]$ located in Germany" is a straightforward adaptation of this framework to GeoCIR.

[D-CIR-2] Baldrati, Agnolucci, Bertini, Del Bimbo. "Zero-Shot Composed Image Retrieval with Textual Inversion." ICCV 2023 / TPAMI 2025. [SEARLE/iSEARLE] Extends Pic2Word with an optimization-based distillation step: slow per-image textual inversion (Gal et al. 2022) generates high-quality pseudo-tokens offline, then a fast feed-forward network is distilled from these tokens. Adds a GPT-2 regularization loss to keep pseudo-tokens in-distribution relative to natural language. Also introduces the CIRCO benchmark (multiple ground truths per query) and the first quantitative analysis of what pseudo-tokens actually encode. Directly relevant because iSEARLE's regularization strategy can guide the training of P for GeoCIR.

[D-CIR-3] Gu, Chun, Kim, Kang, Yun. "Language-Only Training of Zero-Shot Composed Image Retrieval." CVPR 2024. [LinCIR] The most efficient textual inversion method to date. Eliminates the image encoder from the training phase entirely by introducing Self-Masking Projection: a network trained to predict masked spans from CLIP text captions, which generalizes to images via CLIP's alignment. Trains in 48 minutes on 8 A100 GPUs. For GeoCIR, LinCIR's language-only training scheme is particularly compelling: geographic modification texts are textually rich, and a language-only pre-training on geographic caption pairs ("a photo of a landmark in Germany") could efficiently learn a projection that already encodes geographic context without requiring image-GPS triplets.

[D-CIR-4] Tang, Cai, Lin, Nie, de Melo, Zhang. "Context-I2W: Mapping Images to Context-Dependent Words for Accurate Zero-Shot Composed Image Retrieval." AAAI 2024. Addresses

a limitation of Pic2Word: the same image maps to the same pseudo-token regardless of the text modifier, which can be suboptimal when different modifiers highlight different visual attributes. Context-I2W introduces an Intent View Selector that produces context-dependent pseudo-tokens — different projections of the same image depending on what the text modifier implies should change. For GeoCIR, this is highly relevant: the projection of the Tokyo Skytree image should emphasize structural/architectural features (for a geographic query) rather than photographic style or lighting.

[D-CIR-5] Song, Lin, et al. "Fine-grained Textual Inversion Network for Zero-Shot Composed Image Retrieval." SIGIR 2024. [FTI4CIR] Decomposes the single pseudo-token into subject-oriented and attribute-oriented pseudo-tokens, mapped jointly to the text embedding space using a BLIP-generated caption template for alignment. This fine-grained decomposition addresses the information bottleneck in single-token projection: a complex landmark image may need multiple pseudo-tokens to capture both the overall structure and the specific architectural details. For GeoCIR retrieval of structurally similar but geographically distinct landmarks, the attribute-level tokens could encode architectural features (lattice structure, spherical observation deck) that are key for matching across geographic contexts.

[D-CIR-6] Hou, Lin, Wen, Liu, Song. "KEDs: Knowledge-Enhanced Dual-stream Zero-Shot Composed Image Retrieval." IJCNN 2025. Adds a second stream that aligns pseudo-word tokens with named textual concepts mined from image-text pairs, beyond the visual-textual alignment in Pic2Word. For GeoCIR, this dual-stream approach could be adapted to specifically align geographic pseudo-tokens with named geographic concept words ("tower," "spire," "antenna mast"), giving the model an explicit vocabulary for architectural category matching across countries.

Section 2 — Theoretical foundations

[D-TH-1] Gal, Alaluf, Atzmon, Patashnik, Bermano, Chechik, Cohen-Or. "An Image is Worth One Word: Personalizing Text-to-Image Generation using Textual Inversion." ICLR 2023. The paper that introduced the term and concept of textual inversion. Given 3–5 images of a concept, the method optimizes a continuous embedding vector v^* in the CLIP text token space that best reconstructs those images when used as a pseudo-word in Stable Diffusion prompts. This is the direct predecessor to Pic2Word and SEARLE: instead of slow per-image optimization, CIR methods distill this process into a fast feed-forward projection network. Understanding Gal et al.'s optimization objective, the reconstruction loss and the mechanism by which v^* is injected into the text encoder's input layer, is essential for understanding what Formulation D's projection P is being trained to accomplish.

[D-TH-2] Cohen, Gal, Meirom, Chechik, Atzmon. "This is My Unicorn, Fluffy: Personalizing Frozen Vision-Language Representations." ECCV 2022. [PALAVRA] The first paper to apply textual inversion for image retrieval rather than generation, using CLIP directly. PALAVRA finds a pseudo-word v^* that represents a personalized concept within CLIP's text embedding space, then uses CLIP similarity for retrieval of that concept in new contexts. This is architecturally

identical to Formulation D, applied to personal objects rather than geographic contexts. Pic2Word explicitly cites PALAVRA as its direct predecessor.

[D-TH-3] Radford et al. CLIP (ICML 2021). Formulation D depends critically on CLIP's text encoder being receptive to injected continuous vectors that were not part of its original vocabulary. CLIP was trained on 400M image-text pairs with contrastive alignment — this training creates a smooth, well-structured text token embedding space where nearby vectors correspond to semantically similar concepts. The success of textual inversion depends on this smoothness: a pseudo-token that lands "near" the embedding of "Tokyo Skytree" should retrieve images of Tokyo Skytree-like structures.

[D-TH-4] Li, Li, Xiong, Hoi. "BLIP: Bootstrapping Language-Image Pre-training for Unified Vision-Language Understanding and Generation." ICML 2022. BLIP provides the image captioning component used in SEARLE and FTI4CIR to generate semantic regularization targets for pseudo-token training. In the GeoCIR setting, BLIP (or BLIP-2) can be used to generate geographic captions for geotagged training images from YFCC100M, providing the caption-based supervision signal for training the projection P in Formulation D without manually annotated geographic triplets.

[D-TH-5] Gu, Chun, Kim, Kang, Yun. LinCIR (CVPR 2024) — also listed under [D-CIR-3]. Deserves a second mention here as theoretical foundation because its core theoretical contribution — that a language-only projection trained on text self-prediction transfers to images via CLIP's alignment — is a theorem-level claim about the structure of CLIP space. It implies that the geographic projection P in Formulation D can be trained purely on text descriptions of geographic scenes without needing paired image data, which dramatically reduces the annotation burden for GeoCIR.

Cross-cutting references

These papers span multiple formulations and should be treated as background reading for all four.

[X-1] Song, Lin, Wen, Hou, Xu, Nie. "A Comprehensive Survey on Composed Image Retrieval." TOIS 2025. arXiv:2502.18495. Covers 120+ papers with a unified taxonomy and experimental comparison. The most thorough single source for understanding how all composition strategies relate to each other.

[X-2] Du, Deng, Li, Li, Tian. "A Survey on Composed Image Retrieval." TOMM 2025. Second comprehensive survey, with slightly different taxonomy emphasis and more recent results.

[X-3] Vivanco Cepeda, Nayak, Shah. GeoCLIP (NeurIPS 2023). The enabling work for all GeoCIR formulations. Required reading regardless of which formulation you pursue.

[X-4] Baldrati et al. "CIRCO: Composed Image Retrieval on Common Objects in Context." ICCV 2023. The most rigorous CIR benchmark, with multiple ground truths per query and mAP@K evaluation. Any GeoCIR evaluation protocol should be designed with CIRCO's methodology as a template.

[X-5] Liu, Bhatt, Zhu, Teney, Kankanhalli. "CIRR: Image Retrieval on Real-life Images with Pre-trained Vision-and-Language Models." ICCV 2021. Introduced the standard open-domain CIR benchmark with the $R_{\text{subset}@K}$ discriminative metric. Essential reading for understanding how CIR evaluation is typically structured.